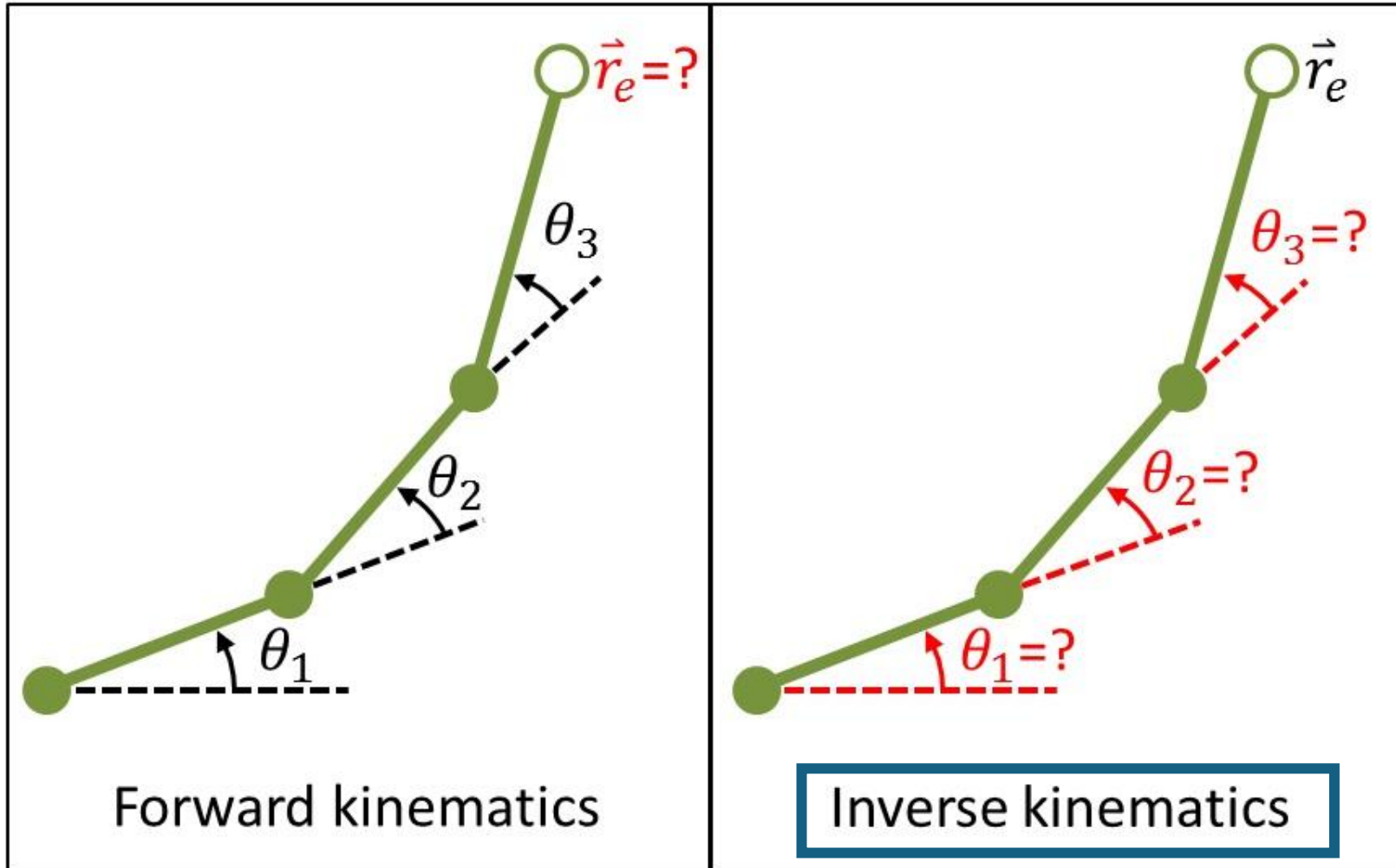


A Novice Version of Inverse Kinematics

Hacker: Haengbok Chung

Remind: inverse kinematics (IK)



DH-parameter: simple initialization

Robotics101 from VisCircuit

Forward Kinematics Inverse Kinematics Dynamics Control Theory Motion Planning Math Tools

Modified Denavit-Hartenberg parameters
(Modified DH Parameters)

Here is the table of modified DH parameters. Feel free to try.

Link	α_{n-1}	a_{n-1}	θ_n	d_n	type	axis
1	0	0	0	10	Revolute	☒
2	0	15	0	10	Revolute	☒
3	41	0	-9	15	Revolute	☒
4	-36	-11	0	10	Revolute	☒

End Effector Position: x: 5.05, y: -15.98, z: 40.36

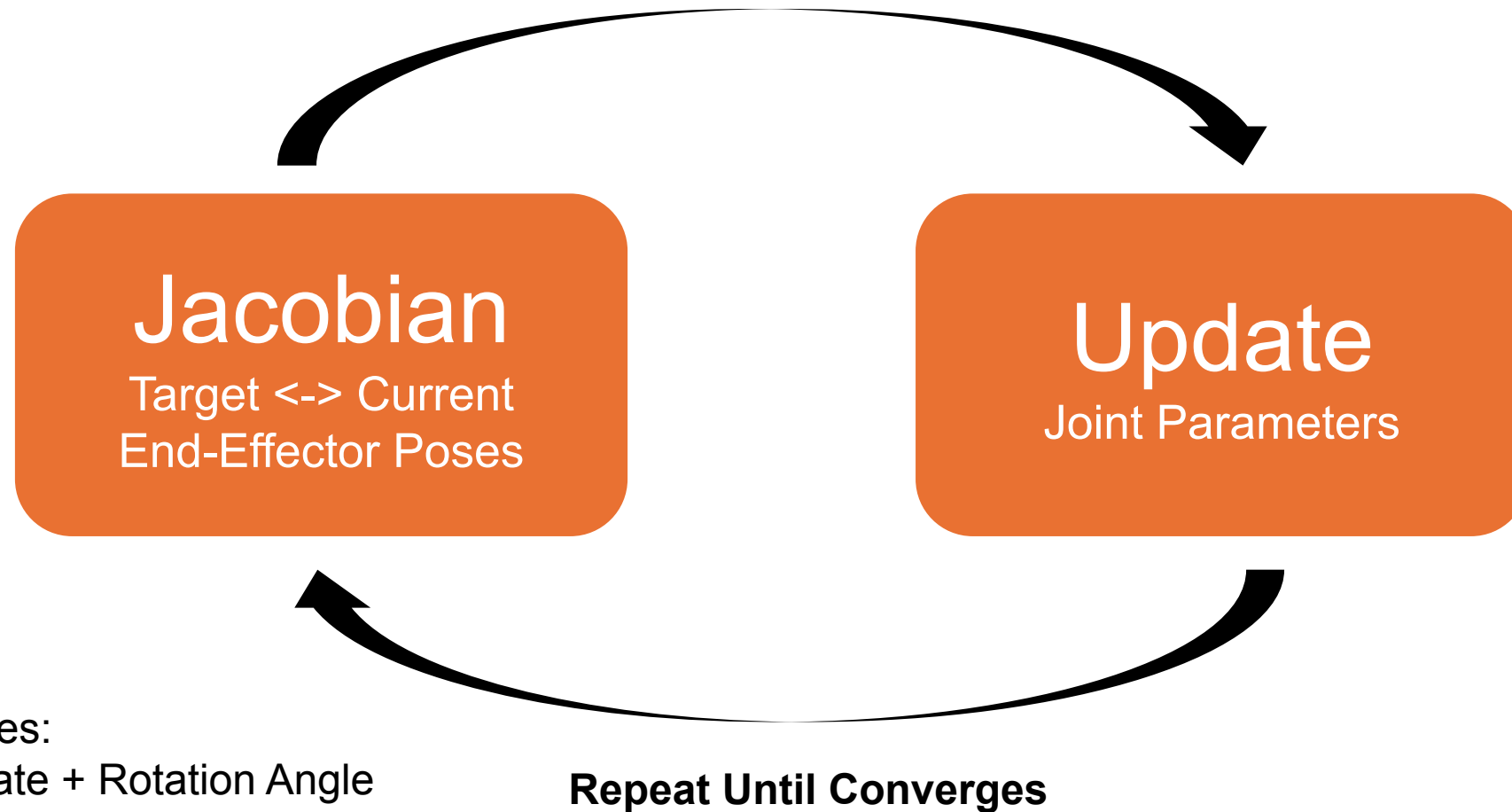
The unit of θ_n and α_n are degree.
The unit of d and a are cm.

Revolute Joint : d, α, a are set during the initial setup. θ changes during operation..
Prismatic Joint : θ, α, a are set during the initial setup. d changes during operation..

The difference between the classic DH parameters and the

Only need rotation angle, distance of Z and X axis

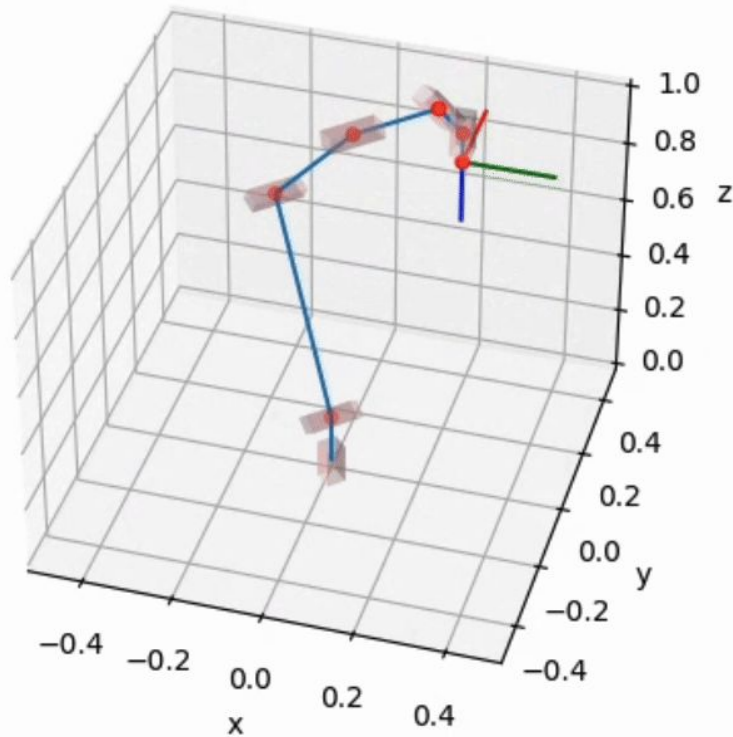
How IK works? (numerical method)



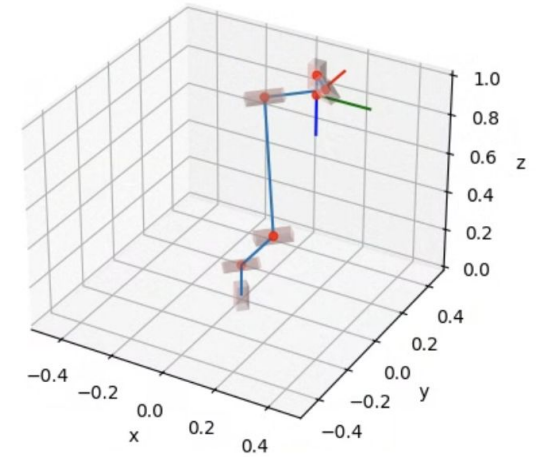
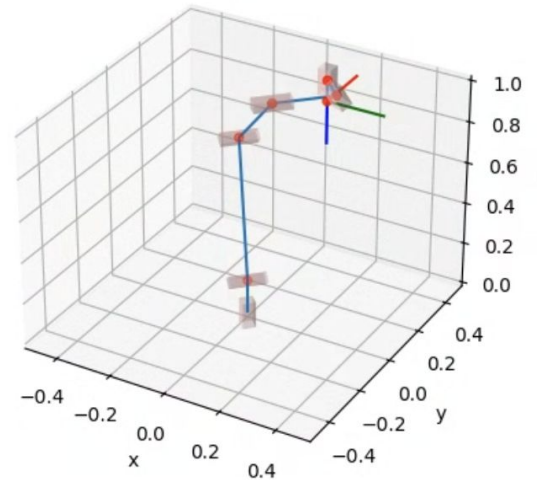
* Pose includes:
XYZ Coordinate + Rotation Angle

Result 1- single arm

Single Arm



**Multiple IK
solutions can exist!**

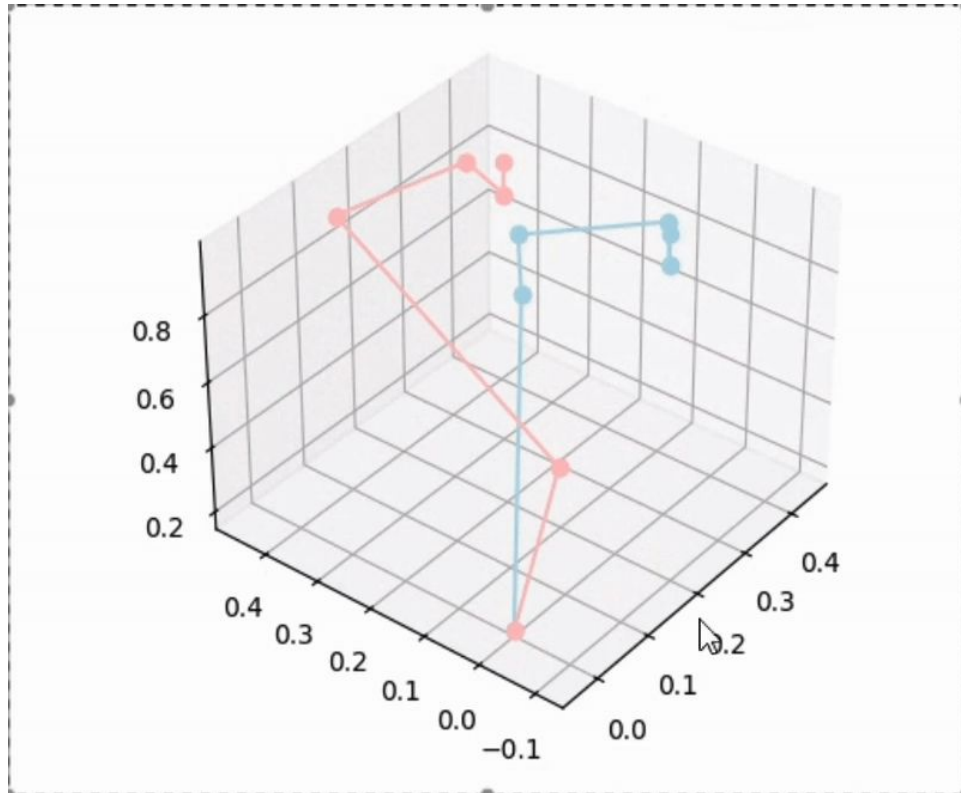


Target end-effector poses are the same

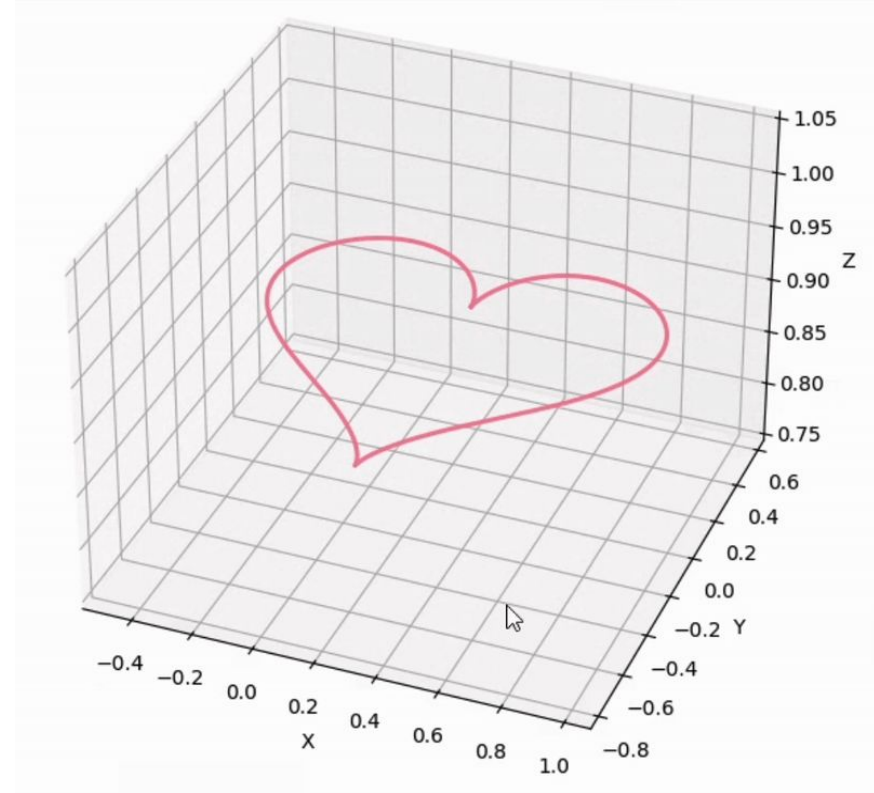
* Libraries: visual_kinematics, mlp_toolkits, matplotlib, ect

Result 2 – variations

Dual Arms

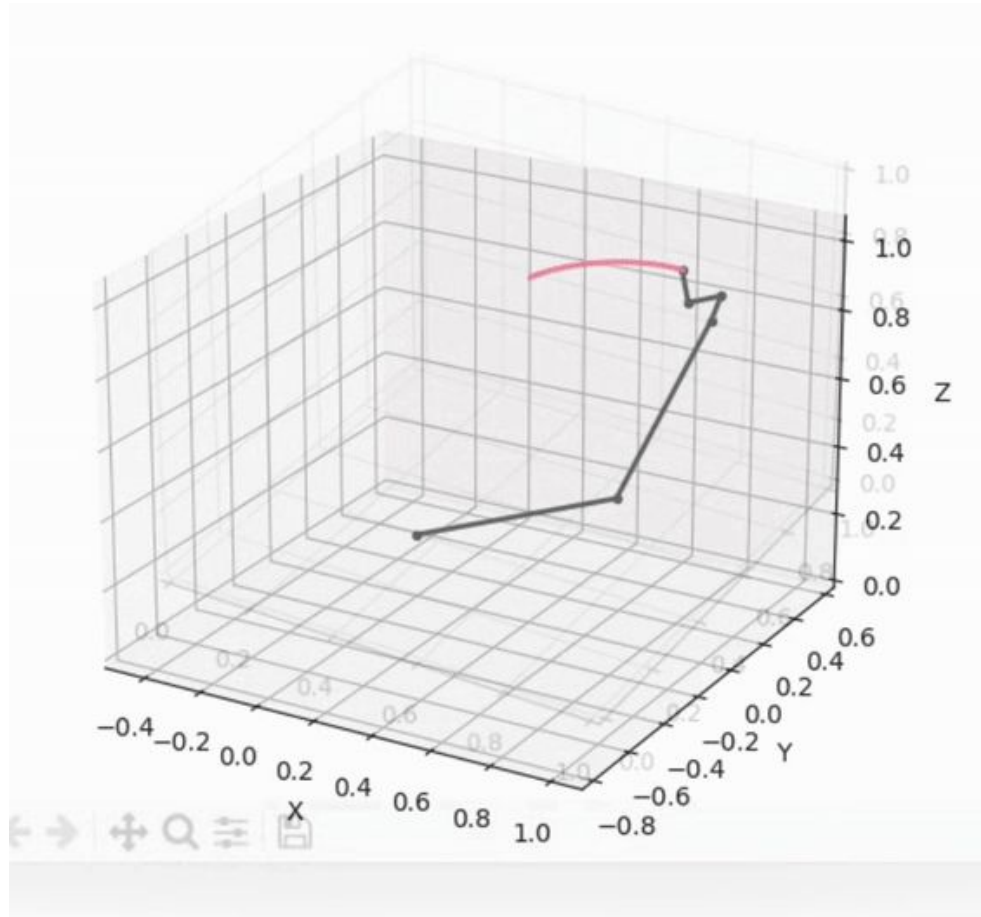


Various Shape (Heart)

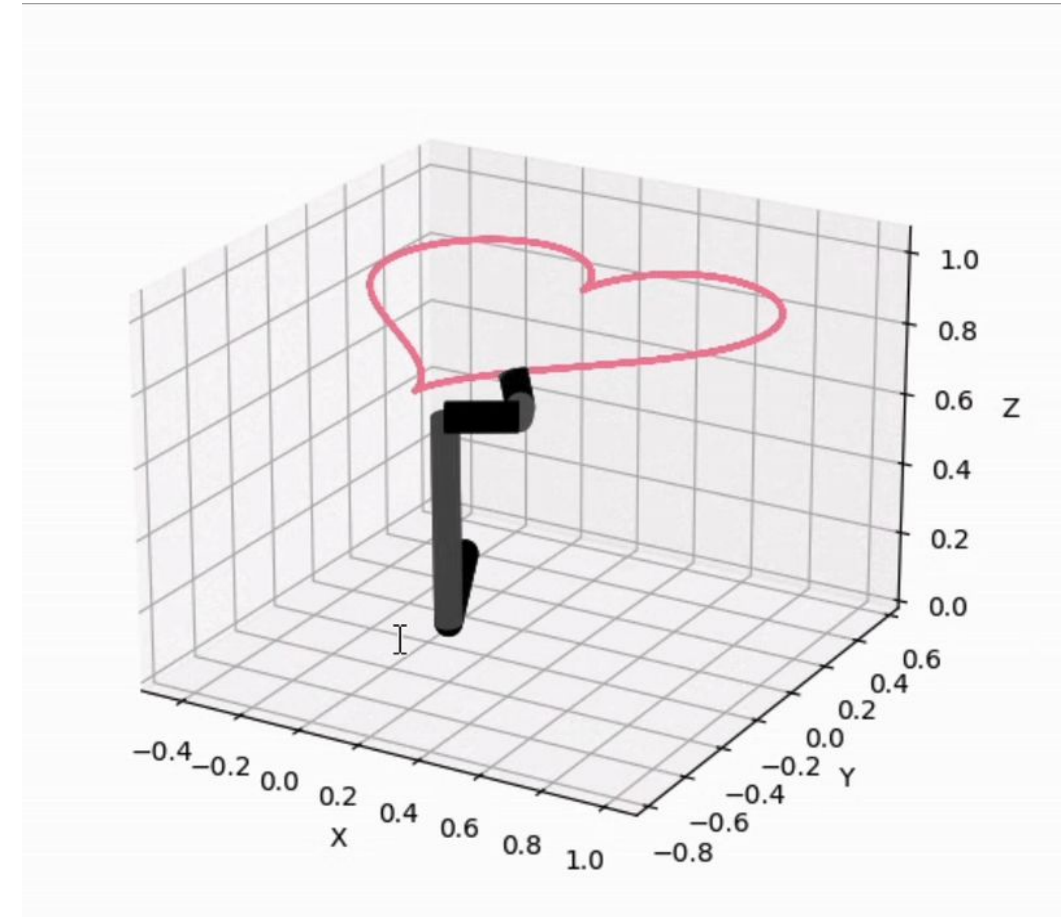


Resut 3 - animated trajectory

Moving Arm Drawing Heart



Moving Arm Drawing Heart with Simple Mesh



Lessons, Limitations

- This is interesting! I'm sincerely grateful for people who advised me for this Hacker project
- Although it also seemed very challenging, learning step by step from easier one made it more enjoyable
- However, this version is not the human-like shaped robot
- It can only performs pre-defined actions

Take home messages

- Inverse Kinematics: pose $\rightarrow$ parameters
- Numerical method is generalizable to complex robot structures
- "Numerical method + AI" hybrid approach is also used
- Although it is very primitive and humble approach, there are tons of amazing, fancy challenges, and works out there!

Disclosure of AI usage

I used Chat-GPT (version 4o) for this assignment.

Access link: <https://chatgpt.com/>

The tool was used several times in September 2025.

Initially, I used Chat-GPT to understand the code.

Then I used AI to draft the code of functions.

All AI-generated content was thoroughly reviewed and revised by me to ensure its accuracy and relevance.

I can provide the unedited transcripts with prompts, interactions.

I take full responsibility for the final work, ensuring that it is my own.